

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0397

Firstenergy Springdale Power Station 138 kV
January 28, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0397. The Interconnection Customer proposes a 140.6 MW solar photovoltaic facility interconnecting at the Firstenergy Springdale Power Station 138 kV substation in the Pennsylvania area. The request was received on October 23, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 230 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,809,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The North Waverly to Firstenergy Springdale Power Station 138 kV is overloaded to 177.5% of its 200 MVA emergency rating for the loss of the Allegheny Energy Units 3 4 & 5 to Southeast Vicksburg 500 kV. The project contributes 29.1% DFAX (40.9 MW) to this overload.

For the loss of Southeast Vicksburg to Allegheny Energy Units 3 4 & 5 500 kV, the North Waverly to Grove 138 kV loads to 171.1% of its 206 MVA rating. Project DFAX: 29.1%.

Pacolet Diesel Generation Facility to Grove Substation 138 kV L: 155.2% loading (219 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 to Southeast Vicksburg 500 kV Line; DFAX = 29.1%.

For the loss of Youngwood to Woodland Tap 345 kV, the Thermas to Wilson 1&2 138 kV loads to 140.6% of its 200 MVA rating. Project DFAX: 7.0%.

Chase Hill to Emily 3 138 kV line: 112.9% loading (162 MVA) following the outage of the Wilson 1&2 to Burwell 345 kV line; DFAX = 10.2%.

Placer to Stinson 138 kV: 112.2% loading (256 MVA) following the outage of the Allegheny Energy Units 3 4 & 5 to Southeast Vicksburg 500 kV; DFAX = 15.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Waverly to Firstenergy Springdale Power Station 138 kV	Rebuild line with higher-capacity conductor	\$2,985,000	100% (\$2,985,000)
North Waverly to Grove 138 kV	Rebuild line with higher-capacity conductor	\$6,308,000	100% (\$6,308,000)
Pacolet Diesel Generation Facility to Grove 138 kV	Rebuild line with higher-capacity conductor	\$6,534,000	100% (\$6,534,000)
Thermas to Wilson 1&2 138 kV	Rebuild line with higher-capacity conductor	\$5,570,000	100% (\$5,570,000)
Chase Hill to Emily 3 138 kV line	Rebuild line with higher-capacity conductor	\$3,521,000	100% (\$3,521,000)
Placer to Stinson 138 kV line	Rebuild line with higher-capacity conductor	\$1,891,000	100% (\$1,891,000)
	Total allocated to project		\$26,809,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.